Radial Transformer for Large-Scale Outdoor LiDAR Point Cloud Semantic Segmentation

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***Abstract*— Semantic segmentation of large-scale outdoor point cloud captured by light detection and ranging (LiDAR) sensors can provide fine-grain and stereoscopic comprehension for the surrounding environment. However, limited by the receptive field of convolution kernel and ignoration of specific spatial properties inherent to the large-scale outdoor point cloud, the existing advanced LiDAR semantic segmentation methods inevitably abandon the unique radial long-range topological relationships. To this end, from the LiDAR perspective, we propose a novel Radial Transformer that can naturally and efficiently exploit the radial long-range dependencies exclusive to the outdoor point cloud for accurate LiDAR semantic segmentation. Specifically, we first develop a radial window partition to generate a series of candidate point sequences and then construct the long-range interactions among the densely continuous point sequences by the self-attention mechanism. Moreover, considering the varying-distance distribution of point cloud in 3-D space, a spatial-adaptive position encoding is particularly designed to elaborate the relative position. Furthermore, we fusion radial balanced attention for a better structure representation of real-world scenes and distant points. Extensive experiments demonstrate the effectiveness and superiority of our method, which achieves 67.5% and 77.7% mean intersection-over-union (mIoU) on two recognized large-scale outdoor LiDAR point cloud datasets SemanticKITTI and nuScenes, respectively.**

***Index Terms*— LiDAR semantic segmentation, long-range features, point cloud, Transformer.**

I. Introduction

LIGHT detection and ranging (LiDAR) sensor can provide point cloud data to accurately describe outdoor scene structures [1], and is the utmost important data source for environmental perception in various intelligent domains, such as intelligent robotics, autonomous driving and remote sensing [2], [3], [4]. Based on the raw LiDAR data, point cloud semantic segmentation can predict semantic category label for each point to achieve fine-grained and stereoscopic perception [5]. The outdoor LiDAR point cloud is typical large-scale, and its accurate semantic-level perception results can offer the fundamental guarantee for various downstream tasks such as path planning and behavior decision-making. Thus, large-scale outdoor LiDAR point cloud semantic segmentation serves as the essentially foundational tasks for modern intelligent domains [6], [7].

However, due to the properties of disorder, sparsity and varying density of point cloud, it poses a great challenge to the data representation for accurate large-scale outdoor LiDAR point cloud semantic segmentation. To address this problem, many recently prevailing methods translate point cloud into regular data structure as pixels for projection-view representation [8], [9] or voxels for voxel-view representation [10], [11]. Benefiting from the powerful learning capability of convolution neural network (CNN), the features behind these structured data representations can be effectively extracted [12], [13]. However, the convolutional kernels of CNN are typically fixed-size square or cubic shape, as shown in Fig. 1. We note that both 2-D convolution and 3-D convolution can only offer limited receptive fields to focus on local features with neighboring points, and it inevitably destroys the native long-range topological relationships between continuous points in 3-D space.

Some recent researches demonstrate that parsing data following its inherent distribution properties can explore a more descriptive representation [14], [15]. Motivated by this thought, we discovery that radial long-range dependencies with a long receptive filed can better preserve the unique radial topologies of point cloud, as shown in Fig. 2. It can be seen that, compared with the convolution kernel, the receptive filed of radial long-range dependencies includes more points. More importantly, the radial long-range dependencies keep certain spatial correlations between continuous points over long distance. In the Radial 1 of Fig. 2, the changes between points are relatively smooth, typically for not only the large-scale classes such as road and vegetation, but also the parts of objects such as the roof of vehicles. While in Radial 2 of Fig. 2, it exhibits abrupt spatial changes when obstacle target appears, especially in the radial and height direction. These spatial changes mainly occur at the edges of objects and small-scale objects. Meantime, we also observe that this radial feature maintains relatively stable (compared with Radial 1 and Radial 3 in Fig. 2), that is, similar radial long-range features can be generalized to other radial directions. Based on the above observations, we seek to utilize the meaningful radial long-range dependencies to provide the reliable and favorable clues for deep learning networks, achieving

Received 14 April 2024; revised 21 July 2024 and 26 September 2024; accepted 1 November 2024. Date of publication 5 November 2024; date of current version 15 November 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFC3002603, in part by the National Natural Science Foundation of China under Grant 62473099, and in part by the Primary Research and Development Plan of Jiangsu Province under Grant BE2022053-5. *(Corresponding author: Xu Li.)*

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Digital Object Identifier 10.1109/TGRS.2024.3492008

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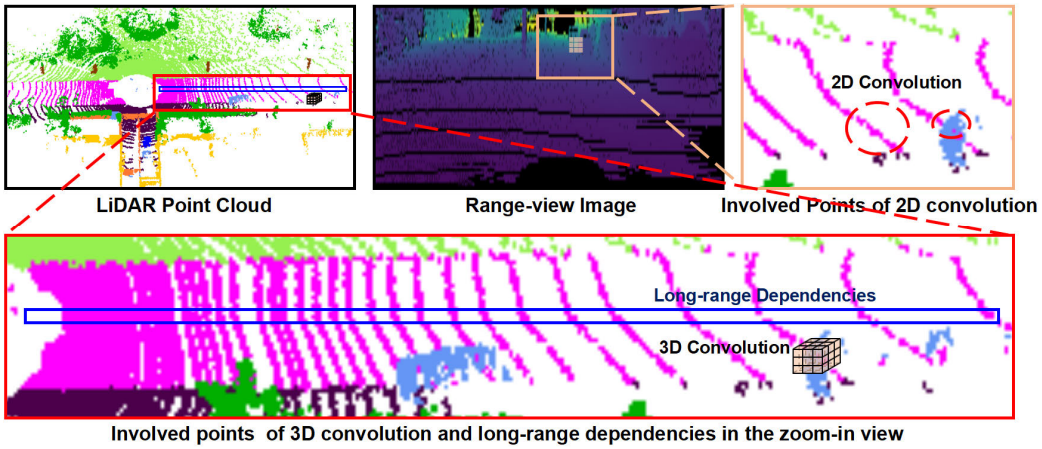


Fig. 1. Receptive field comparisons of 2-D convolutional kernel, 3-D convolutional kernel, and radial long-range dependencies for point cloud.

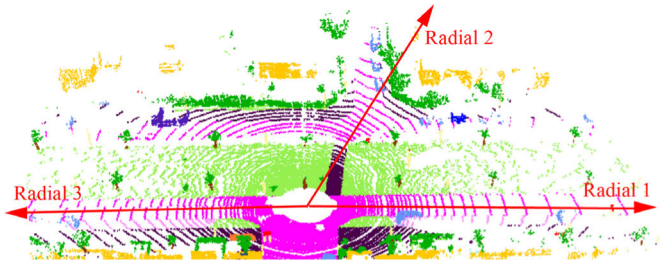


Fig. 2. Long-range dependencies of the point cloud.

accurate large-scale outdoor LiDAR point cloud semantic segmentation.

The challenge is how to construct the radial long-range dependencies we observed from LiDAR scans by deep learning methods. The CNN architecture and its variants are the vital processing approaches for outdoor point cloud segmentation [16], [17]. But CNN-based method benefits for the ability in local neighboring feature extraction, especially for data in a dense and regular format [18]. For exploring radial long-range dependence from varying-density point cloud, CNN-based architecture is obviously unsuitable.

Besides, several recent works seek to explore long-range topologies for large-scale point cloud semantic segmentation. SPG [19] captures super point feature aggregated by neighboring points and constructs the long-range contexts by graph convolutional network. AF-GCN [20] utilizes high-level encoding features to explore the long-range relationships by self-attention mechanism. MCTNet [21] employs cross-attention mechanism to construct long-range interactions between high-level and low-level features for large-scale point cloud semantic segmentation. However, these methods exploit the long-range topologies based on local aggregated superpoint features or high-level features, which limits their ability to perceive detailed information. Our proposed method directly constructs long-range dependencies on point cloud, which can obtain a distinctive spatial description and capture fine-grain details. It is also worth noting that, different with previous long-range features, our method focuses on the radial trait of LiDAR point cloud, which is followed by LiDAR rotating collecting principle and observation perspective.

Transformer is another deep learning approach which is excelled in modeling long-range dependencies by the self-attention mechanism. It is originally designed for the solution for sequence tasks in the domain of natural language processing (NLP), such as translation [22] and text generation [23]. Recently, some researches demonstrate that adaptive applications of Transformer can also yield promising semantic segmentation performance on the images [24], [25], [26] and point clouds domain [27], [28], [29]. But most of these methods based on Transformer are still devoted to exploring local neighboring features of the data and ignore the unique radial long-range spatial topologies of LiDAR point cloud.

To better explore the observed radial long-range dependencies of point cloud, we propose a novel outdoor large-scale LiDAR point cloud semantic segmentation framework dubbed Radial Transformer. Specifically, followed by the rotating collection principle of the LiDAR, we partition point cloud into series of radial windows based on the azimuth angle. Then, we construct the radial long-range dependencies by a self-attention mechanism with hierarchical structure, which the model can learn not only radial long-range features but also local transversal contexts in the deeper layers.

Moreover, to cope with the varying-distance distribution of point cloud in 3-D space, we design a spatial-adaptive position encoding to incorporate a clear expression of relative position for long-range relationship understanding. Furthermore, considering the uneven distribution along the radial direction, radial balanced attention is proposed to improve the model to understand the real-world overall structure and distant points.

In total, our main contributions can be summarized in four folds as follows.

1) We propose an innovative Radial Transformer to exploit the unique radial long-range dependencies of point cloud from LiDAR perceptive by self-attention within the radial windows, which can offer a more detailed and descriptive representation for large-scale outdoor LiDAR point cloud semantic segmentation.
2) An improved spatial-adaptive position encoding method is developed to adapt to 3-D point cloud space with anisotropic distribution, which entitles a more accurate

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relative position information to enhance the radial representation for topological structure of point cloud.
3) We design a radial balanced attention to alleviate the attention bias for close points, improving the comprehension of real-world scene structure and distant points.
4) We conduct extensive quantitative and qualitative experiments on two recognized large-scale outdoor LiDAR point cloud datasets. The results demonstrate that our method can achieve competitive performance compared to the state-of-the-art (SOTA) outdoor LiDAR point semantic segmentation methods.

The rest of this article is organized as follows. Section II reviews the background work related to the topics in this article. Section III describes our proposed method in detail. Section IV shows extensive comparative experiments and evaluation analysis. Finally, we conclude the work in this article and provide prospects for future work in Section V.

II. Related Work

The CNN-based point cloud semantic segmentation, point cloud long-range dependencies, and Transformer are critical background topics of this article, and we revisit the relevant methods in this section.

A. CNN-Based Point Cloud Semantic Segmentation

With the advancement of CNN-based deep learning methods, significant progress has been achieved in outdoor point cloud semantic segmentation. Given the intrinsic properties of disorder, sparsity, and varying density of point cloud, it should be converted into regular representations for facilitating procession. According to LiDAR representations, CNN-based point cloud semantic segmentation approaches can be roughly classified into projection-view method and voxel-view method.

1) *Projection-View Method:* It mainly projects the point cloud into bird's eye view (BEV) image or range view (RV) image by the top–down view or spherical view. SqueezeSeg [30] and SqueezeSegV3 [31] both use the RV image as input to the CNN model and achieve real-time segmentation. RangeNet++ [32] uses the K-nearest neighbor searching method for postprocessing to refine the prediction outputs. CENet [33] integrates the larger convolution kernels and multiple auxiliary segmentation heads for a concise and efficient segmentation. Polarnet [34] quantizes the point cloud into a polar BEV image for a balanced LiDAR representation and employs ring-connected convolution to abstract features. AMVNet [35] aggregates independent BEV- and RV-based features to refine the uncertainty points by late fusion.
2) *Voxel-View Method:* A voxel-based method transforms point cloud into 3-D tiny regular regions for CNN models to perform feature extraction. VoxelNet [36] divides the point cloud into equally spaced 3-D cubic voxels and captures voxel-level feature for LiDAR point cloud semantic segmentation. SparseConv [37] proposes a sparse convolution to only perform the convolution operation on the region with active input points, which greatly improves both efficiency and accuracy. Cylinder3D [15] proposes a 3-D cylindrical partition adaptive to point cloud structure and designs an asymmetrical 3-D convolution network to explore the 3-D geometric pattern.

However, the aforementioned CNN-based methods mainly focus on the neighboring spatial relationships and ignore the unique radial long-range feature exclusive to the outdoor LiDAR point cloud. Thus, in this article, we seek to overcome the representation limitation and explore the innovative and meaningful radial long-range dependencies for outdoor LiDAR point cloud semantic segmentation.

B. Point Cloud Long-Range Dependencies

Point cloud long-range dependencies seek to exploit the large-scale spatial relationships to better understand the environment by point cloud distribution and have been demonstrated to achieve promising performances in point cloud semantic segmentation task.

PointNet [38] utilizes MLP to learn point-wise features and implement the max-pooling operation to abstract global feature for small-scale point cloud classification, part segmentation, and indoor semantic segmentation. SPG [19] formulates a global energy model to geometrically partition point cloud and employs PointNet to aggregate the partitioned points for superpoint feature embedding. Then, it constructs superpoint graph convolution to establish long-range interactions for point cloud semantic segmentation. AF-GCN [20] employs graph convolution to explore neighboring context and adopts the self-attention mechanism on high-level graph neighboring features to model long-range contexts from distant points. MCTNet [21] applies KPConv [39] as a baseline and utilizes a cross-attention mechanism to construct the high-level long-range context spanning various scales for large-scale point cloud semantic segmentation.

These above methods demonstrate that the long-range dependencies are beneficial for point cloud tasks. However, they concentrate on modeling long-range dependencies based on local aggregated features or high-level features, which is difficult to provide the detailed long-range structural topological cues. Thus, how to construct efficient and detailed long-range dependencies focusing on unique radial feature of outdoor LiDAR point cloud with sparseness and varying density still remains to be solved.

C. Transformer

Transformer is another prevailing deep learning architecture and is notable for its self-attention mechanism to model long-range dependencies for sequential data in NLP. Recently, Transformer also achieves tremendous success in its adaptive applications in computer vision and point cloud domain.

1) Vision Transformer: Images are organized in a 2-D pixel format and cannot be directly processed by a standard Transformer in NLP. ViT [40] splits an image into patches and processes them by Transformer for image classification. To achieve pixel-level semantic segmentation, PVT [41] employs a pyramid structure to generate multiscale feature

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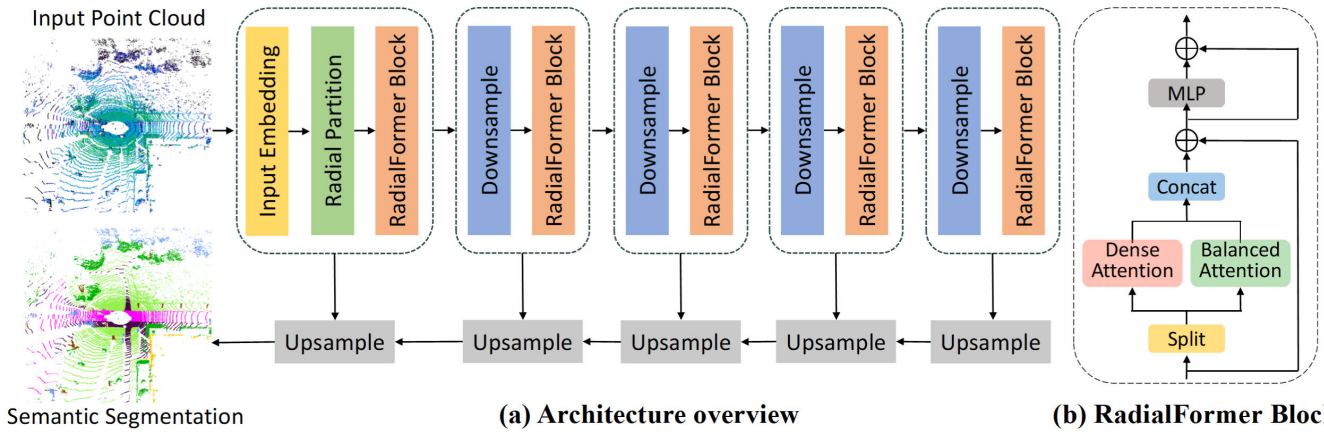


Fig. 3. Architecture of our proposed Radial Transformer. (a) Overview of the network architecture. (b) Structure of the RadialFormer Block. Input point cloud is transformed into higher dimensional features via input embedding module and divided into radial windows through radial partition. Then, the embedded features are fed into RadialFormer block. The downsample layer is to sample points for multiscale feature maps. After several stages combined with RadialFormer block and downsample layer, the encoder features are upsampled for semantic segmentation.

maps for fine-grained predictions. Swin Transformer [42] computes self-attention locally within nonoverlapping window, which can achieve efficient segmentation for high-resolution images.

2) Point Cloud Transformer: Recent works attempt to adopt Transformers for point cloud tasks. VoTr [43] replaces the 3-D convolution with Transformer to construct voxel-level relationships for 3-D object detection. Swformer [45] proposes a window-based Transformer backbone to explore point cloud feature and cross-window correlation to detect 3-D objects. Point Transformer [45] designs a Transformer-based architecture for the local point-level self-attention features with vector attention, which improves upon prior works on semantic scene segmentation. Stratified transformer [46] combines nearby points with distant points to enlarge the effective receptive field for indoor point cloud semantic segmentation. RangeViT [47] exploits the strong patch-wise representation with vision Transformer encoder to benefit outdoor point cloud projection-view semantic segmentation task.

These above Transformer-based methods are adaptively modified to focus on tasks strongly related to local pattern, including computer vision, 3-D object detection, as well as indoor and projection-view outdoor point cloud semantic segmentation. However, relying on local attention or patch-wise attention is challenging to realize accurate fine-gain segmentation for large-scale outdoor LiDAR point cloud. In this article, we seek to explore unique radial long-range dependencies following outdoor point cloud distribution, which can provide the descriptive and detailed spatial topological cues.

III. Methodology

Based on the principle of rotating LiDAR, we propose a novel Radial Transformer to explore the radial long-range dependencies for large-scale outdoor LiDAR point cloud semantic segmentation. In this section, we present our proposed method in detail.

A. Network Architecture

The overall architecture of network is shown in Fig. 3(a). The U-net encoder–decoder structure composed of multiple stages is adopted. In the first stage of encoder, the input point cloud $X \in \mathbb{R}^{N\times 4}$, where features are the 3-D position coordinates and reflection intensity, is transformed to higher dimensional embedding $\tilde{X} \in \mathbb{R}^{N'\times C}$ by the input embedding module with sparse convolution. Also, C notes the channel number of embedded point cloud feature. Then, the embedded point cloud is divided into a series of radial windows through radial partition. Next, the RadialFormer block is employed to construct the radial long-range dependencies by self-attention mechanism. For the consequential encoder stages, they are combined with the downsample layer and RadialFormer block. For each decoder of the network, it is comprised of the upsample layer with residual submanifold convolution module.

The RadialFormer block is the core module of our network. In this block, we simultaneously consider both detail topological relationships and real-world scene structure. It is designed to include radial dense attention and radial balanced attention submodules, as shown in Fig. 3(b). In particular, the input features $F \in \mathbb{R}^{M\times D}$ are first embedded by a residual submanifold module and then split into $F^{\text{den}} \in \mathbb{R}^{M\times D_{\text{hal}}}$ and $F^{\text{bal}} \in \mathbb{R}^{M\times D_{\text{hal}}}$ two halves, where $D_{\text{hal}} = D/2$ is the feature dimension of each half. For the first half feature, we employ radial dense attention to exploit long-range topological relationships based on the partitioned radial window, and we design a spatial-adaptive position encoding method to improve the expression of relative relation. Furthermore, for another half of the input feature, radial balanced attention is additionally designed to extract features for the balanced point distribution to enhance feature expression for real-word scenes and distant points. Finally, two attention outputs are fused to construct the radial long-range feature. The radial partition and RadialFormer block are the core components of our proposed methods, which we provide a more detailed presentation in Sections III-B–III-E.

B. Radial Partition for Point Cloud

LiDAR sensor is mounted on the top of the vehicle and emits a column of fixed number of laser beams (e.g., 32, 64, and 128). The inclination angles of these beams are configured with a specified array to collect environmental information at varying distances from the ego vehicle. Based on the

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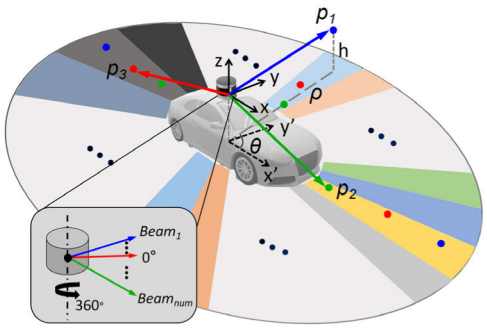


Fig. 4. Radial partition for point cloud based on the LiDAR rotating principle.

column of rotating laser beams, we propose to partition the LiDAR point cloud into radial windows, as shown in Fig. 4. Specifically, we transform the point cloud from the Cartesian coordinate system to cylindrical coordinate system. For each point i, its cylindrical coordinate (ρ_i, θ_i, z_i) can be formulated as follows:

$$\rho_i = \sqrt{x_i^2 + y_i^2}, \quad \theta_i = \arctan \frac{y_i}{x_i} \tag{1}$$

where x_i, y_i, and z_i are the coordinate value of point i in the Cartesian coordinate system.

Then, we calculate the indices for radial windows of point cloud based on the azimuth angles. The index r for the point i marking the radial window can be computed as

$$r = \left\lfloor \frac{\theta_i}{\Delta\theta} \right\rfloor \tag{2}$$

where $\Delta\theta$ denotes the parameter of window size along the azimuth direction. The points whose computed indices are equal are assigned to the same window. By this way, the points are partitioned into multiple equally sized radial windows. In each radial window, the point set includes sequentially consecutive points with different distances. The total number of windows r_{total} can be calculated as

$$r_{\text{total}} = \left\lfloor \frac{2\pi}{\Delta\theta} \right\rfloor. \tag{3}$$

Thereby, the radial window set $R = \{r_1, r_2, \ldots, r_{\text{total}-1}, r_{\text{total}}\}$ for constructing radial long-range dependencies in each point cloud scan is formed.

C. Radial Dense Attention

To extract the dense radial long-range dependencies between points, the Transformer block combined with multihead self-attention is conducted within each partition window, as shown in Fig. 5. Similar to the sequences of token embedding, the sequential point features serve as the input to the Transformer block. For each stage, after the layer normalization, the feature $\tilde{F}_{\text{den}} \in \mathbb{R}^{n\times c}$ is linearly projected into queries $Q \in \mathbb{R}^{n\times c}$, keys $K \in \mathbb{R}^{n\times c}$, and values $V \in \mathbb{R}^{n\times c}$. The linear projection can be expressed as

$$Q = \tilde{F}_{\text{den}} \cdot W_q, \quad K = \tilde{F}_{\text{den}} \cdot W_k, \quad V = \tilde{F}_{\text{den}} \cdot W_v \tag{4}$$

where $W_q, W_k, W_v \in \mathbb{R}^{c\times c}$ are the trainable weight matrices of queries Q, keys K, and values V, respectively. Then,

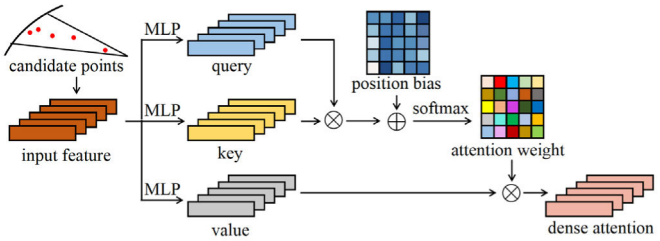


Fig. 5. Radial dense attention.

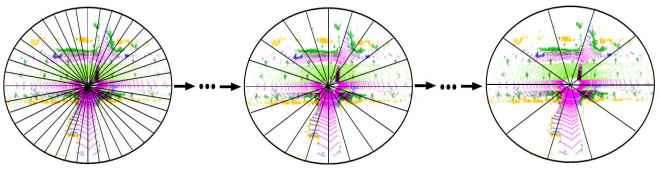


Fig. 6. Enlarged receptive fields with the hierarchical structure as the layer deeper.

the projected features Q, K, and V are split into h heads. To make the Radial Transformer block better understand the spatial topologies, we incorporate the relative position bias $\text{pos_bias}_t \in \mathbb{R}^{n\times n}$ into attention maps followed in [48] and perform the scaled dot-product attention in parallel as follows:

$$\text{att}_t = \text{softmax}\left(\frac{Q_t K_t^T}{\sqrt{d_k}} + \text{pos_bias}_t\right) V_t \tag{5}$$

where $\text{att}_t \in \mathbb{R}^{n\times d}$ denotes the self-attention results of the tth head; $Q_t, K_t, V_t \in \mathbb{R}^{n\times d}$ represent the queries, keys, and values in the tth head; and d_k is the dimension of embedding feature. Finally, the calculated attention results of all heads are concatenated as follows:

$$O = \text{concat}(\text{att}_0, \text{att}_1, \ldots, \text{att}_t, \ldots, \text{att}_{h-2}, \text{att}_{h-1}) \tag{6}$$

where O denotes the output of multihead self-attention for dense distribution.

Meantime, we also consider the local transversal relationships within adjacent radial windows. Inspired by the Swin Transformer [42], which builds hierarchical feature maps by increasing the sizes of image patches, we also generate multiscale features with larger azimuth angles, including local transversal contexts. Different from traditional downsampling operations, we decouple the procession of lowering feature resolution and increasing the receptive field. In particular, to efficiently produce a high-level feature with lower resolution, the feature is first downsampled by 3-D sparse convolution. Then, the radial window size gradually increases to enlarge the receptive field of self-attention for next each stage. As shown in Fig. 6, with the network deeper, the high-level features cover a broader azimuth angle with neighboring radial windows, thus capturing the local transversal features.

D. Spatial-Adaptive Position Encoding

Incorporating position encoding can mitigate the limitations of Transformer-based model in capturing the sequence ordering of token elements, thereby improving the understanding

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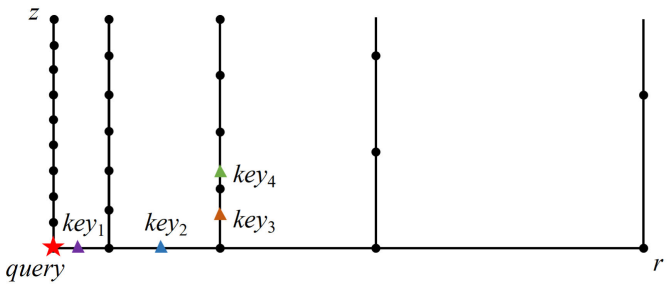


Fig. 7. Adaptive position encoding.

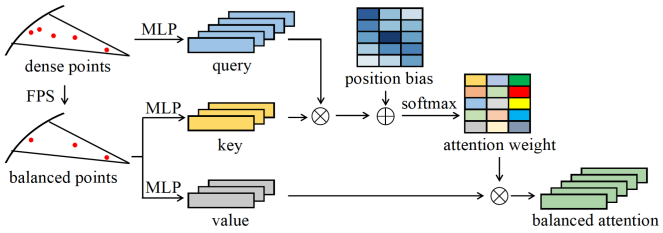


Fig. 8. Radial balanced attention.

of structural topologies [49]. It is noticed that the elements of sentences or images are arranged in a sequential structure. They are evenly spaced and their coordinates naturally provide the distinguishable position coding indices. However, LiDAR point cloud is nonserialized and varying-distance distribution, making it difficult for model to understand the consequential structural relationships by point coordinates. Thus, we combine the absolute and relative position for the position encoding. For LiDAR point cloud data, its input feature actually includes the absolute position coordinates, and we should additionally offer the relative position bias.

However, owing to the inclined emission of LiDAR beams, the gap between adjacent points widens as the distance from LiDAR increases. Besides, the scales of the varying gaps along three axes in the cylindrical coordinate system are inconsistent. These distribution properties bring great challenges for relative position encoding.

To accommodate point cloud spatial distribution, we propose a novel spatial-adaptive position encoding to improve the index expression of the positional relationship between queries and keys, as shown in Fig. 7. In the spatial-adaptive position index of Fig. 7, we follow the varying ruler of spatial gaps between adjacent points, along the radial direction, and the interval between two points grows exponentially as the index increases, while the interval increases linearly along the height direction and keeps constant along the azimuth direction. For query, since we set exponential interval along the radial direction, key_1 and key_2 are entitled to different position indices. If the equal interval is adopted, the nearby points close to LiDAR are assigned to the same index. Besides, for indices far away from the LiDAR, it will produce a large number of empty indices containing no points. Likewise, along the height direction, we set intervals with linear increases, and key_3 and key_4 are given to different indices. By this way, the relative positions can be articulated by indices in point cloud space. For the token j, the indices $(\text{idx}_j^\rho, \text{idx}_j^\theta, \text{idx}_j^z)$ can be formulated as

$$\text{idx}_j^\rho = \max\left(0, \left\lceil \log_2\left(\frac{\rho_j}{\alpha}\right)\right\rceil\right), \quad \text{idx}_j^\theta = \left\lfloor \frac{\theta_j}{\beta} \right\rfloor$$
$$\text{idx}_j^z = \left\lfloor \frac{z_j}{\gamma} \times \rho_j \right\rfloor \tag{7}$$

where α, β, and γ are the starting intervals along three axes. To obtain the relative indices between queries and keys, we then convert them into the nonnegative integer value, which is calculated as

$$\text{idx}_{i,j} = \text{idx}_i - \text{idx}_j + \frac{\text{len}}{2}, \quad j \in \{\rho, \theta, z\} \tag{8}$$

where len is the length of the learnable lookup tables responding to the relative indices. According to the indices $\text{idx}_{i,j}^\rho$, $\text{idx}_{i,j}^\theta$, and $\text{idx}_{i,j}^z$, the encoding position $p_{i,j}^\rho$, $p_{i,j}^\theta$, $p_{i,j}^z \in \mathbb{R}^{h\times d}$ can be found in the retrieved tables and sum them up to get the position encoding $p \in \mathbb{R}^{h\times d}$. Then, the responding position bias value $\text{pos_bias}_{t,i,j}$ can be calculated as

$$p = p_{i,j}^\rho + p_{i,j}^\theta + p_{i,j}^z \tag{9}$$
$$\text{pos_bias}_{t,i,j} = Q_{t,i} \cdot p_t^T + K_{t,j} \cdot p_t^T \tag{10}$$

where $p_t \in \mathbb{R}^d$ is tth position encoding feature, $Q_{t,i} \in \mathbb{R}^d$ is the ith query feature of tth head, and $K_{t,j} \in \mathbb{R}^d$ is the jth key feature of the tth head. Then, the position bias matrix of the tth head $\text{pos_bias}_t \in \mathbb{R}^{n\times n}$ can be obtained and the radial dense attention output can be calculated by (5).

The proposed spatial-adaptive position encoding method discriminatively constructs the relative relationships between the queries and keys based on their distributions along axes in the cylindrical coordinate system. It can prevent that the close points are too dense to be assigned to a single index and avoid the meaningless index assignment for no points in distant sparser region. Combining the absolute coordinates and the relative indices with learnable table, the model can enhance the contextual relationships for the radial long-range dependencies of point cloud.

E. Radial Balanced Attention

In the radial window, the points are exhibited in the long-tailed distribution which the close points are denser and the distant points are much sparser. However, the presence of the objects in the real-world scenes is independent of the density of the point cloud. Therefore, it is difficult to reflect the true sizes and positional topologies of the targets through the relationships of point cloud. In addition, the uneven distribution also brings unfair feature attention to the model, leading to a significant degradation in the segmentation performance as the distance increases. To improve the realistic representations of the scenes and the segmentation result of the distant point, we propose a radial balanced attention submodule to entitle even attention to the points with different distances, as shown in Fig. 8.

Specifically, we downsample the points by the farthest point sampling (FPS) strategy [50] within each radial window and obtain the point cloud with a relatively balanced distribution

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over a long range. Given the point set $P_{\text{in}} = \{p_1, p_2, \ldots, p_n\}$ within the constructed radial window, the most distant points are first chosen as the initial point. Consequently, we iteratively calculate and update the distance between the selected point set and the rest points and select the most distant point to form the target point subset $P_{\text{out}} = \{p_1, p_2, \ldots, p_m\}$. The FPS operation can be expressed as

$$\text{SR} = \frac{n}{m} \tag{11}$$

$$P_{\text{out}} = \text{FPS}(P_{\text{in}}, \text{SR}) \tag{12}$$

where n is the number of input points P_{in}, m represents the number of sampled points P_{out}, and SR is the radio of the sampling. Compared with dense point cloud, the FPS sampling operation can efficiently maintain the distant points and remain a balanced distribution. Moreover, these sampled key points can also provide a better understanding of the topological structure reflecting the real-world scenes.

After sampling, we model the balanced long-range relationship by the cross self-attention. Different from standard Transformer, the key and the value are linearly projected from the feature of sampled points sets $\tilde{F}_{\text{bal}} \in \mathbb{R}^{m\times c}$, while the query is mapped from dense points $\tilde{F}_{\text{den}} \in \mathbb{R}^{n\times c}$ to maintain the resolution. The features Q', K', and V' can be represented as

$$Q' = \tilde{F}_{\text{den}} \cdot W'_q, \quad K' = \tilde{F}_{\text{bal}} \cdot W'_k, \quad V' = \tilde{F}_{\text{bal}} \cdot W'_v \tag{13}$$

where $W'_q, W'_k, W'_v \in \mathbb{R}^{c\times c}$ are the trainable weight matrices of queries Q', keys K', and values V', respectively. Then, we calculate the balanced attention result att'_t as (5).

Finally, dense attention and balanced attention are concatenated to fuse and form the output feature $\tilde{O}$, which can be expressed as

$$\tilde{O} = \text{concat}\left(\text{att}_0, \text{att}_1, \ldots, \text{att}_{h-1}, \text{att}'_0, \text{att}'_1, \ldots, \text{att}'_{h-1}\right). \tag{14}$$

IV. Experiments

In this section, we conduct extensive experiments on two recognized large-scale outdoor LiDAR point cloud datasets to evaluate the performance of our method. First, we introduce datasets, evaluation metrics, and implementation details. Then, compared with the prior SOTA methods, we show the detailed quantitative and qualitative results of semantic segmentation on the above two datasets. Furthermore, to testify the effectiveness of each designed component of our proposed method, the ablation study is conducted.

A. Datasets

1) SemanticKITTI: SemanticKITTI [51] is the pioneering large-scale outdoor point cloud dataset for LiDAR semantic segmentation, whose scene consists of 22 sequences, including inner city traffic, residential areas, highway scenes, and countryside roads. It provides more than 43 000 scans of the raw point cloud data captured by a 64-beam LiDAR sensor. Specifically, 19 130 scans from sequence 00–07 and sequence 09–10 are training set annotated with 19 semantic classes, and 4071 scans of sequence 08 are used for validation, while the remaining 20 351 scans from sequence 11–21 are employed for online testing without providing publicly semantic labels.

2) nuScenes: nuScenes [52] is another widely used dataset for outdoor LiDAR point cloud semantic segmentation, covering a broad range of 1000 driving scenes of totally 40 157 scans of point cloud data collected by 32-beam LiDAR senor. Similar to SemanticKITTI, its samples are split by three sets, with 28 130 samples of 700 scenes for training labeled by 16 categories, 6019 samples of 150 scenes for validation, and another 6008 samples of 150 scenes are selected for online testing.

B. Evaluation Metrics

To evaluate the semantic segmentation results, two metrics, intersection over union (IoU) and mean IoU (mIoU), are introduced, which are widely applied in semantic segmentation task. Specifically, for the lth class, the IoU score is to calculate the proportional relationship between the intersection set and union set of ground truth and predicted labels, which employs to evaluate the predictive performance of model for each class. Also, the metric mIoU is to evaluate the ability of model to predict for all classes. These two metrics can be calculated as

$$\text{IoU}_l = \frac{\text{TP}_l}{\text{TP}_l + \text{FP}_l + \text{FN}_l} \tag{15}$$

$$\text{mIoU} = \frac{1}{L}\sum_{l=1}^{L} \frac{\text{TP}_l}{\text{TP}_l + \text{FP}_l + \text{FN}_l} \tag{16}$$

where TP_l denotes the point that is correctly predicted. FP_l means the point that is predicted to the lth class and the ground truth belongs to other classes. FN_l denotes the point labeled with the lth class, but it is predicted to be other classes. L is the number of all classes in the dataset.

C. Implementation Details

The experiments are implemented on four GeForce RTX 3090 GPUs with 24 GB by PyTorch. For the RaidalFormer block, the embedding dimension and the self-attention head of encoder stages are [32, 64, 128, 256, 256] and [2, 4, 8, 16, 16], respectively. The radial window size of the first layer is set to 0.2° and our model is trained for 50 epochs. The learnable parameters are optimized by the Adamw optimizer whose momentum is set to 0.9 and weight decay rate is 0.02. The initial learning rate lr is set to 0.006 and is adjusted by the Poly scheduler where power is 0.9. In the training process, the cross-entropy loss is used to supervise the model, which can be calculated as

$$\text{loss}_{\text{CE}} = -\frac{1}{p}\sum_{i=1}^{p} g_i \log\left(\hat{y}_i\right) \tag{17}$$

where p is the number of points in one batch; for the ith point, g_i is the label of ground truth and $\hat{y}_i$ is the prediction by model.

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TABLE I

SEMANTIC SEGMENTATION RESULTS ON SEMANTICKITTI VALIDATION SET, WHICH ALL SCORES ARE GIVEN IN PERCENTAGE (%). BOLD NUMBERS INDICATE THE BEST RESULTS, AND THE BLUE NUMBERS INDICATE THE SECOND-BEST RESULTS

Method	mIoU	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic-sign
PolarNet[34]	53.6	91.5	30.7	38.8	46.4	24.0	54.1	62.2	0.0	92.4	47.1	78.0	**1.8**	89.1	45.5	85.4	59.6	72.3	58.1	42.2
SalsaNext[9]	59.4	90.5	44.6	49.6	86.3	54.6	74.0	81.4	0.0	93.4	40.6	69.1	0.0	84.6	53.0	83.6	64.3	64.2	54.4	39.8
RangeViT[47]	60.9	94.7	44.1	61.4	71.8	37.7	65.3	75.5	0.0	**95.5**	**48.8**	83.1	0.0	88.3	60.0	86.3	65.3	72.7	63.1	42.7
SPVNAS[16]	62.3	**96.5**	44.8	63.1	59.9	**64.3**	72.0	86.0	0.0	93.9	42.4	75.9	0.0	88.8	59.1	**88.0**	67.5	73.0	63.5	44.3
CENet[33]	63.8	96.2	**54.5**	71.6	85.6	54.1	67.3	82.4	0.0	**95.5**	45.3	**83.5**	0.1	87.5	60.6	85.1	65.1	69.9	**66.2**	40.4
Cylinder3D[15]	64.9	96.4	**61.5**	**78.2**	66.3	**69.8**	**80.8**	**93.3**	0.0	94.9	41.5	78.0	1.4	87.5	50.0	86.7	**72.2**	68.8	63.0	42.1
AMVNet[35]	**65.2**	95.6	48.8	65.4	**88.7**	54.8	70.8	86.2	0.0	**95.5**	**53.9**	**83.2**	0.2	**90.9**	**62.1**	87.9	66.8	**74.2**	**64.7**	**49.3**
Ours	**67.5**	**96.7**	50.6	**76.2**	**93.1**	62.8	**78.2**	**90.5**	**0.1**	94.3	47.9	81.8	**5.2**	**91.6**	**63.9**	**88.1**	**70.4**	**74.7**	64.1	**52.5**

TABLE II

SEMANTIC SEGMENTATION RESULTS ON THE NUSCENES VALIDATION SET, WHICH ALL SCORES ARE GIVEN IN PERCENTAGE (%). BOLD NUMBERS INDICATE THE BEST RESULTS, AND THE BLUE NUMBERS INDICATE THE SECOND-BEST RESULTS

Method	mIoU	barrier	bicycle	bus	car	construction	motorcycle	pedestrian	traffic cone	trailer	truck	driveable	other flat	sidewalk	terrain	manmade	vegetation
RangeNet++[32]	65.5	66.0	21.3	77.2	80.9	30.2	66.8	69.6	52.1	54.2	72.3	94.1	66.6	63.5	70.1	83.1	79.8
PolarNet[34]	71.0	74.7	28.2	85.3	90.9	35.1	77.5	71.3	58.8	57.4	76.1	96.5	71.1	74.7	74.0	87.3	85.7
PCSCNet[10]	72.0	73.3	42.2	87.8	86.1	44.9	82.2	76.1	62.9	49.3	77.3	95.2	66.9	69.5	72.3	83.7	82.5
Salsanext[9]	72.2	74.8	34.1	85.9	88.4	42.2	72.4	72.2	63.1	61.3	76.5	96.0	70.8	71.2	71.5	86.7	84.4
CENet[33]	75.2	77.9	36.3	**92.7**	90.0	42.9	77.9	76.8	61.7	68.3	82.9	96.7	70.2	75.1	74.5	90.6	88.6
RangeViT[47]	75.2	75.5	40.7	88.3	90.1	49.3	79.3	77.2	**66.3**	65.2	80.0	96.4	71.4	73.8	73.8	89.9	87.2
AMVNet[35]	76.1	**79.8**	32.4	82.2	86.4	**62.5**	81.9	75.3	**72.3**	**83.5**	65.1	**97.4**	67.0	**78.8**	74.6	**90.8**	87.9
Cylinder3D[15]	76.1	76.4	40.3	91.2	**93.8**	51.3	78.0	78.9	64.9	62.1	84.4	96.8	71.6	**76.4**	75.4	90.5	87.4
SAT3D[11]	76.7	74.3	**43.9**	91.3	92.8	**53.0**	77.8	79.7	64.8	63.3	**85.7**	96.5	**73.4**	74.7	**77.3**	89.9	88.0
RPVNet[6]	**77.6**	**78.2**	**43.4**	**92.7**	**93.2**	49.0	**85.7**	**80.5**	66.0	66.9	84.0	**96.9**	**73.5**	75.9	**76.0**	90.6	**88.9**
Ours	**77.7**	77.5	41.2	**93.2**	92.9	52.1	**84.6**	**80.6**	64.3	**72.7**	**85.7**	**96.9**	72.1	75.3	74.8	**90.8**	**88.9**

D. Quantitative Comparisons

1) Results on SemanticKITTI: We make comparisons with recent representative outdoor LiDAR point cloud semantic segmentation methods, including projection-view method PolarNet [34], SalsaNext [9], RangeVit [47], CENet [33], and AMVNet [35], as well as voxel-view method SPVNAS [16] and Cylinder3D [15]. The quantitative results on SemanticKITTI are shown in Table I. It can be noted that the overall performance of our proposed method outperforms prior semantic segmentation models and gives a 2.3% mIoU gain. Moreover, for the performance metric IoU, our proposed method ranks within the top two among all methods for 13 out of 19 classes, including dynamic small-scale classes (e.g., motorcycle and bicyclist) and large-scale classes (e.g., vegetation, terrain, and parking). Impressively, the performances in the classes of truck, other ground, and traffic sign are improved by more than 3%. For the class motorcyclist, which is few sample and small scale, its predicted result makes a breakthrough with zero.

2) Results on nuScenes: We further evaluate our model on the nuScenes dataset with advanced methods. The projection-view method RangeNet++ [32], the voxel-view method PCSCNet [10] and SAT3D [11], as well as the fusion-based method RPVNet [6] are additionally included for comparison. It is noted that the point cloud is sparser and scenes are more complex. The experimental results also verify the superiority of our proposed method, as shown in Table II. From the experiment results, the overall metric mIoU of our proposed method surpasses other methods, and its performance is comparable to RPVNet fusing range-view, point-view, and voxel-view representations. For metric IoU, our method also achieves improvements in many classes, with eight classes ranked in the top two, including small-scale objects (e.g., pedestrian and motorcycle), middle-scale objects (e.g., bus, truck, and building), and large-scale objects (e.g., driveable ground and vegetation). The remaining eight classes show the minimal gaps from the best performances.

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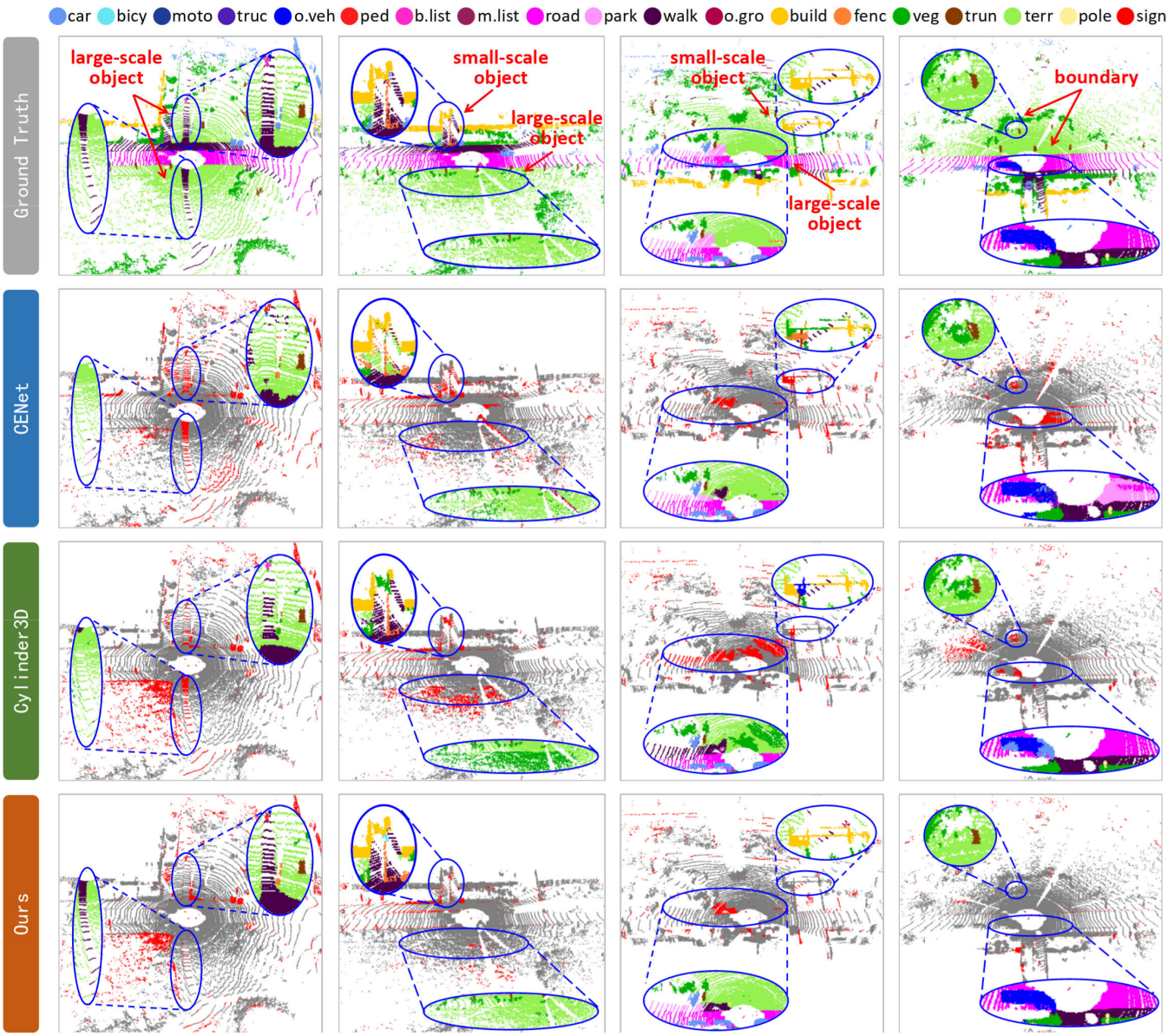


Fig. 9. Qualitative comparison results between SOTA methods and our proposed method on the validation set of SemanticKITTI. The first row shows the ground truth of LiDAR scan, and the last three rows are the difference maps with ground truth (errors shown in red color). The blue cycles show the predicted details with the performance improvement in the zoomed-in view. Compared with CENet and Cylinder3D, our method exhibits lower errors not only in segmenting large-scale objects but also in identifying detailed boundaries and small objects due to radial long-range topological feature.

E. Qualitative Comparisons

1) Results on SemanticKITTI: Fig. 9 visualizes the segmentation results for the advanced projection-view method CENet [33] and voxel-view method Cylinder3D [15] on the SemanticKITTI validation set. As shown in the error maps, we find that the prior methods segment large-scale and distant objects difficult, while our method holistically reduces the false predictions because it effectively models the radial long-range dependencies. Furthermore, we also show the segmentation details with a locally zoomed-in view. It can be seen that our method also generates the finer details and boundaries of the objects.

2) Results on nuScenes: We also compare the qualitative segmentation results with the advanced methods on the nuSences validation set, as shown in Fig. 10. It can be intuitively seen that our proposed method exhibits better segmentation performances than prior methods in large-scale objects and boundary details.

F. Ablation Study

We perform ablation experiments to study the effect of designed elements of the Radial Transformer on the nuScenes validation set and the experimental results are reported in Table III. To better understand the segmentation ability at different distances, we also classify and record the results as close ($\leq$20 m), medium ($>$20 m and $\leq$50 m), and far ($\geq$50 m). From experiments I and II, the radial dense attention significantly improves the segmentation results by 1.48%

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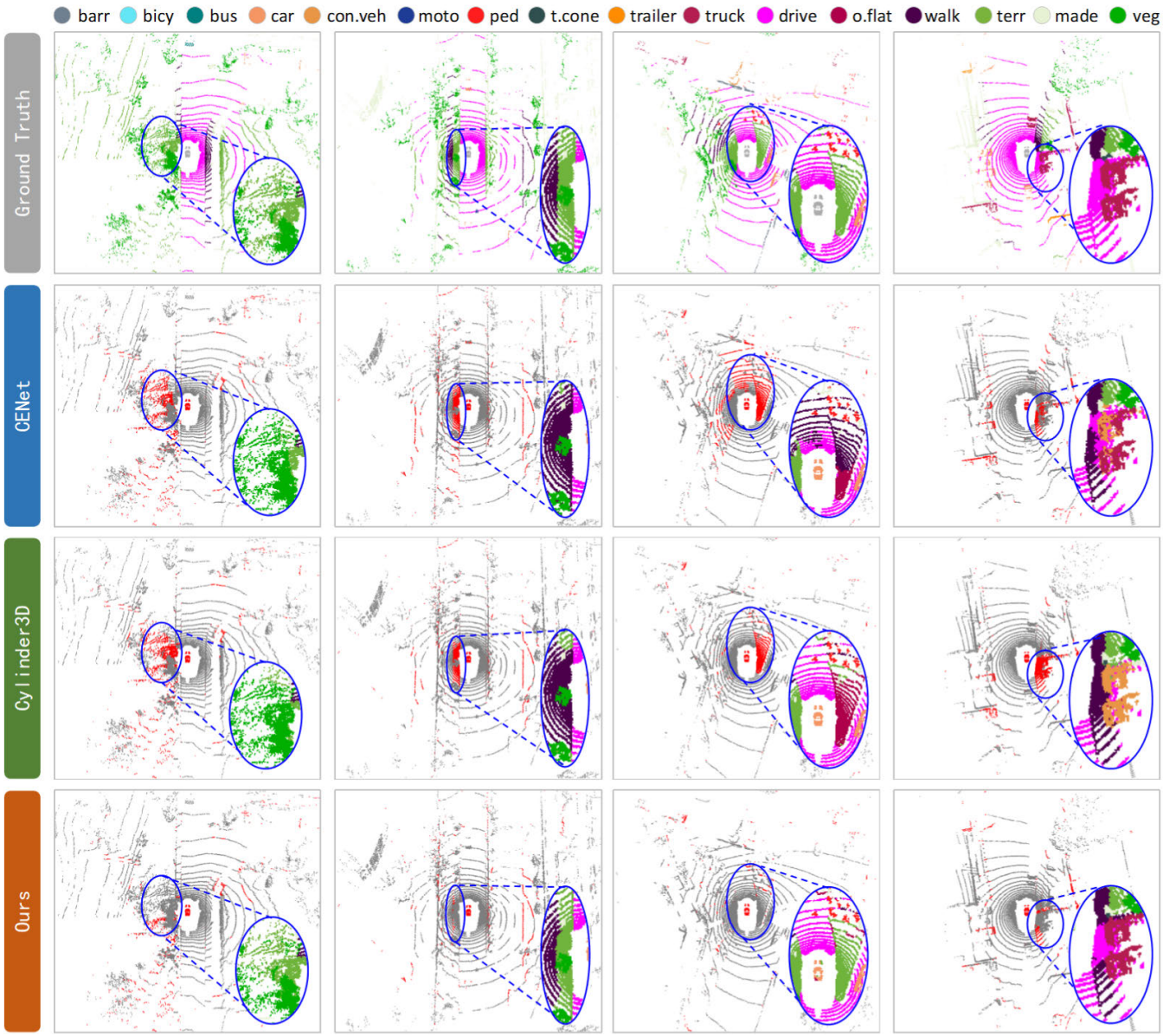


Fig. 10. Qualitative comparison results between SOTA methods and our proposed method on the validation set of nuScenes. The first row shows the ground truth of LiDAR scan, and the last three rows are the difference maps with ground truth (errors shown in red color). The blue cycles show the predicted details with the performance improvement in the zoomed-in view. Compared with CENet and Cylinder3D, our method exhibits lower errors not only in segmenting large-scale objects but also in identifying detailed boundaries.

compared to the baseline SparseConv [37]. It is worth noting that the performance in medium and distant points achieve over 6% and 12% improvement, respectively. The comparison between experiments II and III shows that spatial-adaptive position encoding contributes positively to radial balance attention and improves the performance. Besides, experiments II and IV manifest that incorporating radial balanced attention can elevate the segmentation results in medium and distant distance, and exert minimal performance improvements in the close distance. Moreover, by comparing experiments IV and V, it also reveals that spatial-adaptive position encoding can further enhance the overall performance. The ablation experiments validate that the designed elements are beneficial for outdoor LiDAR point cloud semantic segmentation.

TABLE III

ABLATION STUDY ON THE NUSCENES VALIDATION SET, WHICH ALL SCORES ARE GIVEN IN PERCENTAGE (%) (RDA = RADIAL DENSE ATTENTION, SAPC = SPATIAL-ADAPTIVE POSITION CODING, AND RBA = RADIAL BALANCED ATTENTION)

ID	Baseline	RDA	SAPC	RBA	Close	Medium	Far	mIoU	Δ
I	✓				78.79	51.54	13.28	75.21	-
II	✓	✓			79.54	57.58	25.65	76.69	+1.48
III	✓	✓	✓		80.09	58.05	26.90	77.28	+2.07
IV	✓	✓		✓	79.61	58.03	26.55	76.93	+1.72
V	✓	✓	✓	✓	80.50	58.46	27.25	77.73	+2.52

Furthermore, we investigate the effect caused by the key hyperparameter of radial window size. The results are listed

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TABLE IV
COMPARISON WITH DIFFERENT HYPERPARAMETER OF THE SIZE OF DADIAL WINDOW, WHICH ALL SCORES ARE GIVEN IN PERCENTAGE (%)

Radwin size	0.1°	0.2°	0.4°	0.6°	0.8°	1.0°
mIoU	77.07	77.73	77.20	77.34	76.90	76.58

in Table IV. We note that the optimal size of radial window is 0.2°, setting too small or too large results in a degree of performance degradation.

V. CONCLUSION AND FUTURE WORK

In this article, we introduce a novel Radial Transformer to exploit the unique radial long-range dependencies for large-scale outdoor LiDAR point cloud semantic segmentation. Specifically, based on the LiDAR collecting principle and observation perspective, we develop a radial partition to divide point cloud into radial windows. Then, we construct the radial long-range dependencies within each window by self-attention mechanism. Besides, an improved spatial-adaptive position encoding is proposed to express the relative position of point cloud structure. Considering the uneven spatial distributions of LiDAR data, the balanced attention is designed for further fusion to enhance the understanding of overall structure and distant points in real-world scenes. The extensive comparative experiments on two large-scale outdoor LiDAR point cloud datasets validate the superiority of our proposed method, which achieves 67.5% and 77.7% mIoU on SemanticKITTI and nuScenes, respectively.

In future work, we seek to extend our proposed method by applying it to other point cloud tasks and integrating with other advanced representation methods to achieve more accurate LiDAR point cloud semantic segmentation.

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